# HyperFusion: A Computational Approach for Hyperspectral, Multispectral, and Panchromatic Image Fusion

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***Abstract*— Fusing hyperspectral image (HSI) and multispectral image (MSI) of high spatial resolution is typically utilized to obtain HSIs of high spatial resolution. However, the spatial quality of most existing methods is unsatisfactory due to the limited spatial resolution of an MSI. To further improve the spatial resolution of the fused HSI while keeping the spectral information well, we propose a new computational paradigm, named HyperFusion, which simultaneously fuses HSI, MSI, and panchromatic (PAN) image. To achieve this goal, we first establish two data fidelity terms based on a physical observation that HSI and MSI can be treated as degraded versions of the fused HSI. Consequently, the spatial and spectral information from HSI and MSI can be well preserved. To efficiently transfer the spatial details of PAN into the fused HSI while keeping the spectral information well, we further construct a prior constraint from PAN based on the structural similarity. Meanwhile, we impose another low-rank prior constraint on the coefficient matrix to accurately describe the latent characteristics of the HSI with high spatial resolution. By incorporating the aforementioned data fidelity terms and prior constraints, we finally formulate the objective as an optimization problem and utilize the alternative direction multiplier method to solve it efficiently. Comprehensive experiments on simulated and real datasets are carried out to demonstrate the superiority of HyperFusion over other state of the arts in terms of visual quality and quantitative analysis. We also adopt a simulated experiment of vegetation coverage index analysis to verify the effectiveness of HyperFusion in remote sensing applications.**

***Index Terms*— Alternating direction method of multipliers, image fusion, low rank, sharpening, structural similarity.**

## I. Introduction

HYPERSPECTRAL imaging technique that can accurately identify substances of the scene greatly expands the capabilities of remote sensing technology and plays an important role in the fields of geological survey, crop monitoring, and national defense security [1]–[4]. When the number of bands increases, the photons emitted by the sun scattered on each band of a hyperspectral image (HSI) are correspondingly reduced due to the limited solar irradiance. Therefore, the spatial resolution of an HSI is often low. Compared with an HSI, a multispectral image (MSI) has fewer bands but higher spatial resolution. To obtain an HSI of a high spatial and spectral resolution, fusion of HSI and MSI (FHM) is a typical strategy [5]–[8].

In recent years, FHM methods can be roughly divided into three categories: pansharpening method-based extension, model-based methods, and deep convolutional neural network (CNN)-based methods [9]. Since pansharpening methods [10]–[12] that fuse a panchromatic (PAN) image and an MSI have been developed maturely, there are some attempts with existing pansharpening methods to adapt them for the generalization of fusing HSI and MSI. For example, Chen *et al.* [13] proposed a solution by dividing the spectrum of an HSI into several regions and used conventional pansharpening technology to fuse HSI and MSI in each spectral region. Model-based methods impose certain prior terms to regularize the to-be-estimated high-resolution HSI and estimate the HSI by optimizing the fusion model [14]. These methods establish fusion models, including fidelity and regularization terms based on spatial (spectral) degradation mechanisms and prior information, and utilize effective solutions to solve these models. Consequently, a good balance between spatial and spectral quality can be achieved in model-based methods, attracting considerable research interests in recent years. Dong *et al.* [15] proposed a nonnegative structured sparse representation approach for FHM, which was formulated as a joint estimation of spectral basis and sparse coefficients with the prior knowledge about spatial–spectral sparsity of the HSI. In recent years, some model-based methods build their models from the perspective of tensor and utilize some tensor decomposition methods [16], [17]. Benefiting from the development of CNN [18]–[20], deep CNN-based methods have achieved remarkable achievements. For example, Dian *et al.* [21] directly learned the image prior through residual learning based on deep CNN and then integrated the learned prior into the HSI and MSI fusion framework. Although deep learning fusion frameworks demonstrate their advantages, a large number of images are needed to train specialized deep learning models for different kinds of sensors,

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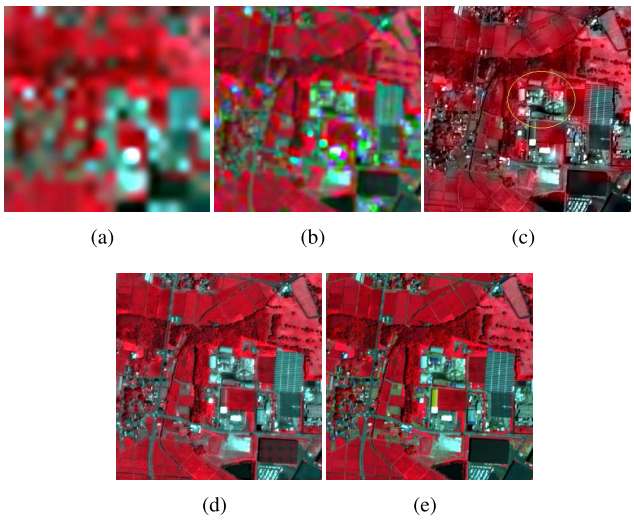



Fig. 1. Example of the HSI fusion experiment. (a) Upsampled HSI. (b) FHM method. (c) FHP method. (d) Proposed method. (e) Reference. We select three bands of fusion results to form a false-color image for demonstration. The proposed method provides satisfactory spatial details with well spectral information.

thereby sacrificing flexibility. Therefore, deep CNN-based methods are not the focus of this study.

Although a variety of FHM methods have been investigated during the past few decades, generating a fused HSI of a high spatial resolution and good spectral quality is still a challenging problem due to the following reasons.

1) The spatial quality of most existing FHM methods is always unsatisfactory because the spatial resolution of an MSI is still not high. This phenomenon can be observed in Fig. 1(b). Compared with the upsampled HSI, FHM can provide better spatial quality. However, the spatial details of Fig. 1(b) still cannot be clearly observed.
2) Recently, fusion of HSI and PAN (FHP) brings new insight for generating an HSI with high spatial quality [22], [23], as shown in Fig. 1(c). However, FHP always suffers significant spectral distortions. For example, the color of the region labeled by a yellow ellipse in Fig. 1(c) is obviously different from that of the reference HSI in Fig. 1(e).
3) More generally, one hyperspectral satellite (e.g., GF-5) and another satellite with the multispectral and PAN instruments (e.g., GF-1) can capture the same scene with relative close time, demonstrating the possibility and effectiveness for generating an HSI with high spatial and spectral quality by fusing HSI, MSI, and PAN (FHMP) simultaneously. This phenomenon can be observed from the comparison of Fig. 1(b)–(d). The proposed method provides more spatial details than the FHM method, and meanwhile, its spectral information is better preserved than the FHP method, leading to the best balance between the spatial and the spectral quality for the fused HSI.

In this study, we propose a novel computational approach for hyperspectral, multispectral, and PAN image fusion, named HyperFusion. We first establish two data fidelity terms based on the fact that HSI and MSI can be regarded as degraded versions of the fused HSI, to preserve the spatial and spectral information of HSI and MSI, respectively. Furthermore, we construct two prior constraints: one prior constraint explored from PAN based on structural similarity, to efficiently transfer the spatial details of PAN into the fused HSI, and another prior constraint considering that the coefficient matrix [24] decomposed from the fused HSI should be low-rank, to accurately describe latent characteristics of the HSI with a high spatial resolution. Finally, we formulate the objective as a fusion model by incorporating the above data fidelity terms and prior constraints and solve this model efficiently by using the alternative direction multiplier method. The main contributions of this study are summarized as follows.

1) We propose a novel model of simultaneous FHMP, which can greatly improve the spatial quality of the fused HSI. In particular, we design an efficient prior constraint based on the structural similarity between PAN and an ideal MSI. Unlike other FHMP methods, this prior constraint from PAN does not require the spectral response transform vector between PAN and HSI, leading to the good preservation of spectral information while efficiently injecting the high-frequency details of PAN into the fused HSI through the ideal MSI. As the investigated model is still ill-posed, we further add another constraint based on the low-rank property of the coefficient matrix. As far as we know, such prior constraints have seldom been investigated in the FHMP studies.
2) As only simulated experiments are demonstrated in previous FHMP studies, we conduct a real experiment based on the fusion of images from GF-5 and GF-1 satellites, where HyperFusion obtains superior fusion results over other state of the arts. A simulated vegetation coverage experiment is also provided to demonstrate the effectiveness of HyperFusion in remote sensing.

The remainder of this article is organized as follows. Section II introduces the related work. In Section III, we describe HyperFusion in detail. In Section IV, we conduct a lot of fusion experiments, and a vegetation index analysis experiment is further carried out to verify the effectiveness of our method in Section V. The conclusion is reported in Section VI.

## II. Related Work

This section briefly reviews the background material of our work, including FHM, FHP, and FHMP.

### A. FHM Methods

In recent years, a large number of FHM methods have been proposed [25]–[27]. These methods are generally divided into the following three main categories: pansharpening method-based extension, model-based methods, and deep CNN-based methods. The pansharpening method-based extension mainly includes component substitution and multiresolution analysis. Gram–Schmidt adaptive (GSA) [28], [29] is a well-known pansharpening method, which can be directly applied to FHM.

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Selva *et al.* [30] proposed a framework to effectively adapt the multiresolution analysis to the HSI and MSI fusion. For model-based methods, regularization models are built to solve the fusion problem of FHM in accordance with the mechanism of spectral and spatial degradation. Therefore, proper fidelity and regularization terms are important for the optimal solution of the optimization model. Yokoya *et al.* [31] utilized the coupled nonnegative matrix factorization (CNMF)-based alternating spectrum unmixing method to estimate the endmember and abundance matrix. Simoes *et al.* [24] proposed a subspace-based FHM method, which used the endmember extraction technology to obtain the endmember (subspace) matrix from HSI, and then, the prior term was established by introducing the vector total variation regularization based on the piecewise smoothing characteristic of the discontinuous HSI. Since the calculation was carried out in the subspace matrix, the computational cost was reduced. Due to the high correlation among spectral bands of HSI, 3-D images can be naturally represented by 3-D tensors. Consequently, some studies considered solving the problem of FHM from the perspective of tensor [32]–[35]. Li *et al.* [32] proposed a coupled sparse tensor factorization method, which treated an HSI as a 3-D tensor and redefined the fusion problem as the estimation of the core tensor and the dictionary of three modes. Dian *et al.* [36] divided an MSI into multiple blocks and applied the learned clustering structure and coefficients to form a 4-D tensor. Then, the FHM problem was expressed as a low tensor-train rank regularization optimization problem. The deep CNN-based methods construct a deep network and produce the fused HSI by feeding HSI and MSI into the network [37]. Palsson *et al.* [38] designed a 3-D CNN for FHM. Zheng *et al.* [39] introduced the deep residual CNN into the FHM process by mapping the residual between the initialized and the reference HSIs. Xie *et al.* [40] proposed an FHM model and solved it efficiently according to a CNN by unfolding the iterative algorithm.

### B. FHP Methods

The FHP method that fuses HSI and PAN brings new insight for generating an HSI with high spatial quality. This method is mainly derived from pansharpening methods and can be broadly divided into four classes: component substitution, multiresolution analysis, variational methods, and deep CNN-based methods. Component substitution methods replace a component (e.g., a spectral transformation) of an MSI with a PAN. Typical component substitution methods include the matting model [41], nonlinear intensity–hue–saturation [42], and nonlinear PCA [43]. In multiresolution analysis methods, spatial details of a PAN are injected into an MS through a multiscale decomposition of the PAN, such as the generalized Laplacian pyramid (GLP) method [44] and morphological filters [45]. In recent years, variational methods achieve good balances between the spatial and the spectral quality by formulating suitable fusion models. Ballester *et al.* [46] proposed the first variational method, which treated the upsampled MS image as a blur version of the high spatial resolution multispectral (HRMS) image. A dynamic gradient sparsity was further exploited as a prior in [47] to improve spatial quality. Fu *et al.* [48] considered a local gradient constraint in different local patches and bands, leading to satisfactory fusion performance. Similar to FHM, deep CNN-based methods form a new branch in pansharpening [49], [50]. Recent state of the arts included pansharpening by using a CNN [51], deep residual learning [52], [53], and multiscale CNN [54]. However, FHP is more complex than pansharpening because the spectral range of HSI normally is much wider than the one of the corresponding PAN, while MSI and PAN are usually acquired almost in the same spectral range. Therefore, significant spectral distortions are inevitable.

### C. FHMP Methods

As we know, only a few studies have been focused on FHMP methods. Yokoya *et al.* [55] first applied CNMF to FHMP. Bendoumi and Benlefki [56] proposed an unconstrained least-square unmixing scheme for FHMP. Arablouei [57] considered the problem of fusing an arbitrary number of multiband images belonging to the same scene, which casted the fusion problem as a reduced dimension linear inverse problem. In these methods [55]–[57], a spectral response transform vector that represents the transform from the spectral response function of HSI to that of PAN is necessarily required. However, due to the large difference between the spatial resolution of HSI and PAN, the estimation of this spectral response transform vector generated from the HSI with a low spatial resolution and PAN is always inaccurate, leading to some inevitable spectral distortions in the fused HSI.

## III. Proposed Method

In the following, we first introduce some notations of our study. We denote an HSI with a size of $W_h \times H_h \times L_h$ as a matrix $\mathbf{H} \in \mathbb{R}^{L_h \times W_h H_h}$, which means that the HSI has $L_h$ bands and each band of the HSI has $W_h \times H_h$ pixels. An MSI with a size of $W_m \times H_m \times L_m$ and a PAN with a size of $W_p \times H_p$ are denoted as matrices $\mathbf{M} \in \mathbb{R}^{L_m \times W_m H_m}$ and $\mathbf{P} \in \mathbb{R}^{1 \times W_p H_p}$, respectively. $L_m$ represents the number of bands in the MSI. $W_m \times H_m$ and $W_p \times H_p$ are the spatial resolution of MSI and PAN, respectively. We denote the fused HSI of the high spatial and spectral resolution as a matrix $\mathbf{F} \in \mathbb{R}^{L_h \times W_p H_p}$. The resolution ratio between PAN and HSI is $(W_p/W_h)$ and that between PAN and MSI is $(W_p/W_m)$. The diagram of the proposed method is shown in Fig. 2.

### A. Proposed Model

*1) Construction of Data Fidelity Terms:* The HSI $\mathbf{H}$ can be regarded as the spatial degraded version of the fused HSI $\mathbf{F}$ as

$$\mathbf{H} = \mathbf{F}\mathbf{B}_h\mathbf{S}_h. \tag{1}$$

The matrix $\mathbf{B}_h \in \mathbb{R}^{W_p H_p \times W_p H_p}$ is a spatial blur matrix. $\mathbf{S}_h \in \mathbb{R}^{W_p H_p \times W_h H_h}$ is a spatial downsampling operation that downsamples the spatial resolution of the fused HSI into that of the HSI.

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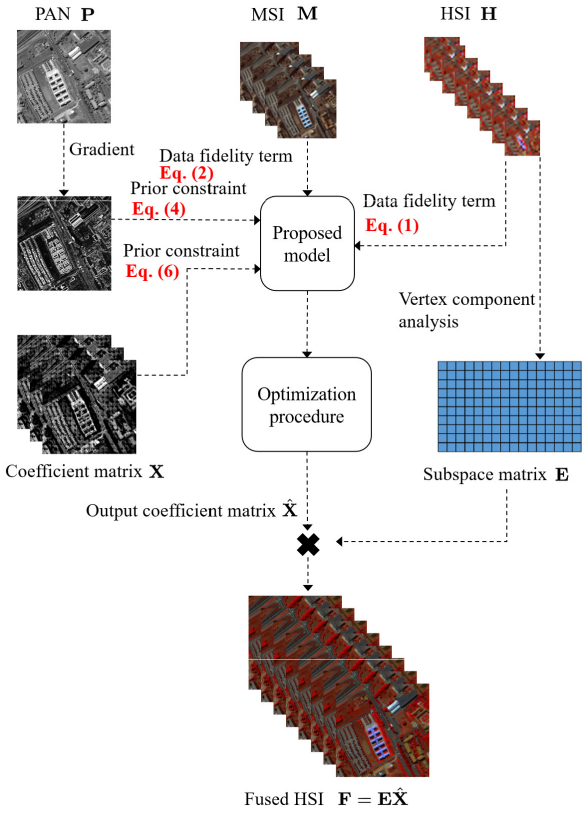



Fig. 2. Diagram of the proposed method.

The MSI $\mathbf{M}$ can be regarded as the degraded version of the fused HSI $\mathbf{F}$ after blurring and spatial–spectral downsampling

$$\mathbf{M} = \mathbf{R}_m \mathbf{F} \mathbf{B}_m \mathbf{S}_m \tag{2}$$

where $\mathbf{R}_m \in \mathbb{R}^{L_m \times L_h}$ is the spectral response matrix of the multispectral instrument. The matrix $\mathbf{B}_m \in \mathbb{R}^{W_p H_p \times W_p H_p}$ is a spatial blur matrix. $\mathbf{S}_m \in \mathbb{R}^{W_p H_p \times W_m H_m}$ is a spatial downsampling operation that downsamples the spatial resolution of the fused HSI into that of the MSI.

*2) Construction of Prior Constraints:* Recovery of the high-dimensional HSI $\mathbf{F}$ from the low-dimensional images $\mathbf{H}$, $\mathbf{M}$, and $\mathbf{P}$ is an ill-posed problem. Therefore, proper priors are necessarily required for accurately recovering $\mathbf{F}$.

First, some previous studies [55]–[57] utilized an estimated spectral response transform vector $\mathbf{R}_p \in \mathbb{R}^{1 \times L_h}$ to construct the relationship between PAN $\mathbf{P}$ and the fused HSI $\mathbf{F}$ as

$$\mathbf{P} = \mathbf{R}_p \mathbf{F}. \tag{3}$$

However, the estimated $\mathbf{R}_p$ from PAN $\mathbf{P}$ and HSI $\mathbf{H}$ is always inaccurate because the resolution ratio between PAN and HSI is large. Consequently, (3) will be not strictly obeyed, leading to some spectral distortions.

To avoid this drawback, we focus on the consistency of the high-pass filtered components of images [48] to improve the spatial resolution while keeping the spectral information well. As all bands are representations of the same objects, their gradients tend to be at the same spatial positions, which means that they should not only be sparse but also group sparse in the gradient domain. Therefore, we first construct a prior based on the above structural similarity [47] between PAN $\mathbf{P}$ and an ideal MSI $\mathbf{R}_m\mathbf{F}$ with the same spatial resolution of PAN as

$$\| \bigtriangledown \bar{\mathbf{P}} - \bigtriangledown \mathbf{R}_m \mathbf{F} \|_{2,1} \tag{4}$$

where $\| \cdot \|_{2,1}$ represents the $l_{2,1}$ norm that encourages group sparsity and $\bigtriangledown$ is the gradient operator. $\bar{\mathbf{P}}$ means duplicating $\mathbf{P}$ into multiple bands.

Second, to reduce the computational cost, some FHM methods [24] decompose the fused HSI $\mathbf{F}$ into the product of a subspace matrix $\mathbf{E} \in \mathbb{R}^{L_h \times D}$ and a coefficient matrix $\mathbf{X} \in \mathbb{R}^{D \times W_p H_p}$. $\mathbf{X}$ denotes the coefficients represented by the linear combination of vector members in the subspace matrix $\mathbf{E}$ for each pixel of $\mathbf{F}$, and $D$ represents the number of vectors of pure spectral features in the subspace obtained by the HSI through image decomposition. Then, the solution of the high-dimensional data was changed to the calculation of the coefficient matrix in the low-dimensional subspace, which efficiently reduced the computational complexity. Based on the above assumption, the following relationship is established:

$$\mathbf{F} = \mathbf{E}\mathbf{X}. \tag{5}$$

Therefore, we add another prior constraint on the coefficient matrix $\mathbf{X}$, rather than $\mathbf{F}$, to reduce the computational cost. Inspired by the method in hyperspectral unmixing [58], we can observe the following phenomena for the coefficient matrix $\mathbf{X}$ decomposed from $\mathbf{F}$.

As the subspace matrix $\mathbf{E}$ is always made of pure endmembers extracted from the HSI, it always has the full column rank property. Therefore, the coefficient matrix $\mathbf{X}$ is a low-rank matrix because it is evident that the fused HSI is always a low-rank matrix. As a result, we can utilize the nuclear norm $\|\mathbf{X}\|_*$ to describe the low-rank property of $\mathbf{X}$ as

$$\|\mathbf{X}\|_* = \text{trace}\left(\sqrt{\mathbf{X}^T\mathbf{X}}\right). \tag{6}$$

Based on (1), (2), and (4)–(6), we formulate the proposed fusion model as

$$\begin{aligned} \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{H} - \mathbf{E}\mathbf{X}\mathbf{B}_h\mathbf{S}_h\|_F^2 + \frac{\lambda_m}{2}\|\mathbf{M} - \mathbf{R}_m\mathbf{E}\mathbf{X}\mathbf{B}_m\mathbf{S}_m\|_F^2 \\ + \lambda_\phi \| \bigtriangledown \bar{\mathbf{P}} - \bigtriangledown \mathbf{R}_m\mathbf{E}\mathbf{X}\|_{2,1} + \lambda_l\|\mathbf{X}\|_* \end{aligned} \tag{7}$$

where $\| \cdot \|_F^2$ represents the Frobenius norm. $\lambda_m$, $\lambda_\phi$, and $\lambda_l$ are parameters to balance different terms. $\mathbf{B}_h$, $\mathbf{B}_m$, and $\mathbf{R}_m$ are estimated from $\mathbf{H}$ and $\mathbf{M}$. More details of calculating these matrices can be found in [24]. $\mathbf{E}$ is obtained from the HSI by the vertex component analysis method [59].

### *B. Optimization Procedure*

We use the alternative direction multiplier method [60] to solve the optimization problem of (7). By introducing five auxiliary variables $\mathbf{O}$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$, and $\mathbf{Q}$, the optimization problem of (7) is expressed as

$$\begin{aligned} \arg \min_{\mathbf{X},\mathbf{O},\mathbf{U},\mathbf{V},\mathbf{W},\mathbf{Q}} & \frac{1}{2}\|\mathbf{H} - \mathbf{E}\mathbf{O}\mathbf{S}_h\|_F^2 + \frac{\lambda_m}{2}\|\mathbf{M} - \mathbf{R}_m\mathbf{E}\mathbf{U}\mathbf{S}_m\|_F^2 \\ & + \lambda_\phi \| \bigtriangledown \bar{\mathbf{P}} - \bigtriangledown \mathbf{W}\|_{2,1} + \lambda_l\|\mathbf{Q}\|_* \\ \text{s.t.,} \quad & \mathbf{O} = \mathbf{X}\mathbf{B}_h, \quad \mathbf{U} = \mathbf{X}\mathbf{B}_m, \quad \mathbf{V} = \mathbf{X} \\ & \mathbf{W} = \mathbf{R}_m\mathbf{E}\mathbf{V}, \quad \mathbf{Q} = \mathbf{X}. \end{aligned} \tag{8}$$

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The augmented Lagrangian function of (8) is expressed as

$$\begin{aligned} L = &\frac{1}{2}\|\mathbf{H}-\mathbf{EOS}_h\|_F^2+\frac{\lambda_m}{2}\|\mathbf{M}-\mathbf{R}_m\mathbf{EUS}_m\|_F^2 \\ &+\lambda_\phi\|\bigtriangledown\bar{\mathbf{P}}-\bigtriangledown\mathbf{W}\|_{2,1}+\lambda_l\|\mathbf{Q}\|_* \\ &+\frac{\mu}{2}\left\|\mathbf{XB}_h-\mathbf{O}-\frac{\mathbf{Y}_1}{\mu}\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{XB}_m-\mathbf{U}-\frac{\mathbf{Y}_2}{\mu}\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{X}-\mathbf{V}-\frac{\mathbf{Y}_3}{\mu}\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{R}_m\mathbf{EV}-\mathbf{W}-\frac{\mathbf{Y}_4}{\mu}\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{X}-\mathbf{Q}-\frac{\mathbf{Y}_5}{\mu}\right\|_F^2 \end{aligned} \tag{9}$$

where $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, $\mathbf{Y}_4$, and $\mathbf{Y}_5$ are Lagrange multipliers and $\mu$ is a penalty parameter. The optimization problem of (9) can be decomposed into seven subproblems, which are solved separately.

*1)* $\mathbf{X}$ *Subproblem:* We solve $\mathbf{X}^{(t+1)}$ from

$$\begin{aligned} \arg\min_{\mathbf{X}} &\left\|\mathbf{XB}_h-\mathbf{O}^{(t)}-\frac{\mathbf{Y}_1^{(t)}}{\mu}\right\|_F^2+\left\|\mathbf{XB}_m-\mathbf{U}^{(t)}-\frac{\mathbf{Y}_2^{(t)}}{\mu}\right\|_F^2 \\ &+\left\|\mathbf{X}-\mathbf{V}^{(t)}-\frac{\mathbf{Y}_3^{(t)}}{\mu}\right\|_F^2+\left\|\mathbf{X}-\mathbf{Q}^{(t)}-\frac{\mathbf{Y}_5^{(t)}}{\mu}\right\|_F^2 \end{aligned} \tag{10}$$

where the superscript $t$ represents the number of iterations. The solution of (10) is

$$\begin{aligned} \mathbf{X}^{(t+1)} = &\left[\left(\mathbf{O}^{(t)}+\frac{\mathbf{Y}_1^{(t)}}{\mu}\right)\mathbf{B}_h^T+\left(\mathbf{U}^{(t)}+\frac{\mathbf{Y}_2^{(t)}}{\mu}\right)\mathbf{B}_m^T\right. \\ &\left.+\left(\mathbf{V}^{(t)}+\frac{\mathbf{Y}_3^{(t)}}{\mu}\right)+\left(\mathbf{Q}^{(t)}+\frac{\mathbf{Y}_5^{(t)}}{\mu}\right)\right]\times\left[\mathbf{B}_h\mathbf{B}_h^T+\mathbf{B}_m\mathbf{B}_m^T+2\mathbf{I}\right]^{-1}. \end{aligned} \tag{11}$$

*2)* $\mathbf{O}$ *Subproblem:* We solve $\mathbf{O}^{(t+1)}$ from

$$\arg\min_{\mathbf{O}}\frac{1}{2}\|\mathbf{H}-\mathbf{EOS}_h\|_F^2+\frac{\mu}{2}\left\|\mathbf{X}^{(t+1)}\mathbf{B}_h-\mathbf{O}-\frac{\mathbf{Y}_1^{(t)}}{\mu}\right\|_F^2. \tag{12}$$

We use the mask matrix $\mathbf{S}_h$ to divide $\mathbf{O}$ into $\mathbf{OS}_h$ and $\mathbf{O}\bar{\mathbf{S}}_h$, where $\bar{\mathbf{S}}_h$ represents the pixel that is not selected by the matrix $\mathbf{S}_h$, and then, the solution of (12) is changed into

$$\begin{cases} \mathbf{O}^{(t+1)}\mathbf{S}_h = [\mathbf{E}^T\mathbf{E}+\mu\mathbf{I}]^{-1} \\ \qquad\qquad\times\left[\mathbf{E}^T\mathbf{H}+\mu\left(\mathbf{X}^{(t+1)}\mathbf{B}_h-\dfrac{\mathbf{Y}_1^{(t)}}{\mu}\right)\right]\mathbf{S}_h & (13) \\ \mathbf{O}^{(t+1)}\bar{\mathbf{S}}_h = \left(\mathbf{X}^{(t+1)}\mathbf{B}_h-\dfrac{\mathbf{Y}_1^{(t)}}{\mu}\right)\bar{\mathbf{S}}_h. & (14) \end{cases}$$

*3)* $\mathbf{U}$ *Subproblem:* We solve $\mathbf{U}^{(t+1)}$ from

$$\arg\min_{\mathbf{U}}\frac{\lambda_m}{2}\|\mathbf{M}-\mathbf{R}_m\mathbf{EUS}_m\|_F^2+\frac{\mu}{2}\left\|\mathbf{X}^{(t+1)}\mathbf{B}_m-\mathbf{U}-\frac{\mathbf{Y}_2^{(t)}}{\mu}\right\|_F^2. \tag{15}$$

Similar to the $\mathbf{O}$ subproblem, we use the mask matrix $\mathbf{S}_m$ to divide $\mathbf{U}$ into $\mathbf{US}_m$ and $\mathbf{U}\bar{\mathbf{S}}_m$, where $\bar{\mathbf{S}}_m$ represents the pixel that is not selected by the matrix $\mathbf{S}_m$. Then, the solution of (15) is

$$\begin{cases} \mathbf{U}^{(t+1)}\mathbf{S}_m = [\lambda_m\mathbf{E}^T\mathbf{R}_m^T\mathbf{R}_m\mathbf{E}+\mu\mathbf{I}]^{-1} \\ \qquad\qquad\times\left[\lambda_m\mathbf{E}^T\mathbf{R}_m^T\mathbf{M}+\mu\left(\mathbf{X}^{(t+1)}\mathbf{B}_m-\dfrac{\mathbf{Y}_2^{(t)}}{\mu}\right)\right]\mathbf{S}_m & (16) \\ \mathbf{U}^{(t+1)}\bar{\mathbf{S}}_m = \left(\mathbf{X}^{(t+1)}\mathbf{B}_m-\dfrac{\mathbf{Y}_2^{(t)}}{\mu}\right)\bar{\mathbf{S}}_m. & (17) \end{cases}$$

*4)* $\mathbf{V}$ *Subproblem:* We solve $\mathbf{V}^{(t+1)}$ from

$$\arg\min_{\mathbf{V}}\frac{\mu}{2}\left\|\mathbf{R}_m\mathbf{EV}-\mathbf{W}^{(t)}-\frac{\mathbf{Y}_4^{(t)}}{\mu}\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{X}^{(t+1)}-\mathbf{V}-\frac{\mathbf{Y}_3^{(t)}}{\mu}\right\|_F^2. \tag{18}$$

The solution of the above problem is

$$\begin{aligned} \mathbf{V}^{(t+1)} = &\left[\mathbf{E}^T\mathbf{R}_m^T\mathbf{R}_m\mathbf{E}+\mathbf{I}\right]^{-1} \\ &\times\left[\mathbf{E}^T\mathbf{R}_m^T\left(\mathbf{W}^{(t)}+\frac{\mathbf{Y}_4^{(t)}}{\mu}\right)+\mathbf{X}^{(t+1)}-\frac{\mathbf{Y}_3^{(t)}}{\mu}\right]. \end{aligned} \tag{19}$$

*5)* $\mathbf{W}$ *Subproblem:* We solve $\mathbf{W}^{(t+1)}$ from

$$\arg\min_{\mathbf{W}}\lambda_\phi\|\bigtriangledown\bar{\mathbf{P}}-\bigtriangledown\mathbf{W}\|_{2,1}+\frac{\mu}{2}\left\|\mathbf{R}_m\mathbf{EV}^{(t+1)}-\mathbf{W}-\frac{\mathbf{Y}_4^{(t)}}{\mu}\right\|_F^2. \tag{20}$$

If $\mathbf{G}=\mathbf{W}-\bar{\mathbf{P}}$, then (20) can be transformed into

$$\begin{aligned} \mathbf{G}^{(t+1)} = \arg\min_{\mathbf{G}}\frac{\mu}{2}\|\mathbf{G}-\left(\mathbf{R}_m\mathbf{EV}^{(t+1)}-\frac{\mathbf{Y}_4^{(t)}}{\mu}-\bar{\mathbf{P}}\right)\|_F^2 \\ +\lambda_\phi\|\bigtriangledown\mathbf{G}\|_{2,1} \end{aligned} \tag{21}$$

which is a typical vectorial total variation denoising problem [61]–[63] and can be directly solved based on the fast iterative shrinkage-thresholding algorithm [64]. Subsequently, we have

$$\mathbf{W}^{(t+1)} = \mathbf{G}^{(t+1)}+\bar{\mathbf{P}}. \tag{22}$$

*6)* $\mathbf{Q}$ *Subproblem:* We solve $\mathbf{Q}^{(t+1)}$ from

$$\arg\min_{\mathbf{Q}}\lambda_l\|\mathbf{Q}\|_*+\frac{\mu}{2}\left\|\mathbf{X}^{(t+1)}-\mathbf{Q}-\frac{\mathbf{Y}_5^{(t)}}{\mu}\right\|_F^2 \tag{23}$$

which can be solved directly by the singular value shrinkage algorithm [65].

*7)* $\mathbf{Y}_1$*,* $\mathbf{Y}_2$*,* $\mathbf{Y}_3$*,* $\mathbf{Y}_4$*, and* $\mathbf{Y}_5$ *Subproblems:* We update $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, $\mathbf{Y}_4$, and $\mathbf{Y}_5$ based on the following formulas:

$$\begin{cases} \mathbf{Y}_1^{(t+1)} = \mathbf{Y}_1^{(t)}-\mu(\mathbf{X}^{(t+1)}\mathbf{B}_h-\mathbf{O}^{(t+1)}) & (24) \\ \mathbf{Y}_2^{(t+1)} = \mathbf{Y}_2^{(t)}-\mu(\mathbf{X}^{(t+1)}\mathbf{B}_m-\mathbf{U}^{(t+1)}) & (25) \\ \mathbf{Y}_3^{(t+1)} = \mathbf{Y}_3^{(t)}-\mu(\mathbf{X}^{(t+1)}-\mathbf{V}^{(t+1)}) & (26) \\ \mathbf{Y}_4^{(t+1)} = \mathbf{Y}_4^{(t)}-\mu(\mathbf{R}_m\mathbf{EV}^{(t+1)}-\mathbf{W}^{(t+1)}) & (27) \\ \mathbf{Y}_5^{(t+1)} = \mathbf{Y}_5^{(t)}-\mu(\mathbf{X}^{(t+1)}-\mathbf{Q}^{(t+1)}). & (28) \end{cases}$$

Finally, the output coefficient matrix $\hat{\mathbf{X}}=\mathbf{X}^{(T_{\max})}$ can be calculated by iteratively solving the above subproblems, and then, the fused HSI $\mathbf{F}$ is derived according to $\mathbf{F}=\mathbf{E}\hat{\mathbf{X}}$. $T_{\max}$ represents the maximum number of iterations.

Based on the above analysis, we summarize the proposed algorithm in Algorithm 1.

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**Algorithm 1:** HyperFusion

**Input** $\mathbf{H}$, $\mathbf{M}$, $\mathbf{P}$, $\mathbf{E}$, $\mathbf{B}_h$, $\mathbf{S}_h$, $\mathbf{B}_m$, $\mathbf{S}_m$, $\mathbf{R}_m$, $\lambda_m$, $\lambda_\phi$, $\lambda_l$, $\mu$, $T_{\max}$.
**Initialization**: $\mathbf{O}$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$, $\mathbf{Q}$, $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, $\mathbf{Y}_4$, and $\mathbf{Y}_5$
**For** $t = 0$ to $T_{\max} - 1$ **do**
  **Update** $\mathbf{X}^{(t+1)}$ in closed-form using (11).
  **Update** $\mathbf{O}^{(t+1)}$ in closed-form according to (13) and (14).
  **Update** $\mathbf{U}^{(t+1)}$ in closed-form using (16) and (17).
  **Update** $\mathbf{V}^{(t+1)}$ in closed-form according to (19).
  **Update** $\mathbf{W}^{(t+1)}$ in closed-form using (21) and (22).
  **Solve** $\mathbf{Q}^{(t+1)}$ from (23) directly by the singular value shrinkage algorithm [65].
  **Update** $\mathbf{Y}_1^{(t+1)}$, $\mathbf{Y}_2^{(t+1)}$, $\mathbf{Y}_3^{(t+1)}$, $\mathbf{Y}_4^{(t+1)}$, and $\mathbf{Y}_5^{(t+1)}$ directly by (24), (25), (26), (27), and (28).
**End**
**Calculate** $\mathbf{F} = \mathbf{E}\hat{\mathbf{X}}$,
**Output** $\mathbf{F}$.

## IV. Experimental Results

### A. Experimental Setting

In the experiment, we adopt three simulated datasets and one real dataset. Simulated datasets are acquired from the public hyperspectral dataset websites and noisy bands are removed already. As for the real dataset, we manually remove some noisy bands and some bands of poor quality. Both the simulated datasets and the real dataset are used at the original radiometric resolution in bits. All datasets are normalized before tested. Simulated datasets, named datasets A, B, and C, are generated from three different flight platforms. Dataset A comes from the HSI of Pavia university taken by the reflective optics system imaging spectrometer (ROSIS) sensor in Italy, which has 103 bands in the wavelength range of 0.43–0.86 $\mu$ m. Dataset B comes from the Salinas HSI taken by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor in America, which has 224 bands in the wavelength range of 0.40–2.45 $\mu$ m. Dataset C is from the Chikusei HSI taken by the Headwall Hyperspec-VNIR-C sensor of an airborne flight in Japan. Its wavelength ranges from 0.36 to 1.02 $\mu$ m with a spectral resolution of 1.29 nm. The spatial resolution of datasets A, B, and C are 1.3, 3.7, and 2.5 m, respectively.

We adopt Wald's protocol [66] to generate the simulated datasets. We take dataset A as an example to detail the simulation process. We first select 93 bands from the original Pavia HSI with the size of $320 \times 320 \times 93$ and use it as the reference (Ref). To generate the HSI with the size of $20 \times 20 \times 93$, we use a modulation transfer function (MTF) filter to blur and downsample Ref for simplicity. Since the gain at the Nyquist frequency is usually unavailable, we choose the default value of 0.3 in our experiment. To simulate the MSI with the size of $80 \times 80 \times 4$, we use the multispectral response of the IKONOS-like reflection spectrum response filter to sample Ref along with the spectral mode and then use an MTF filter (value of 0.15 for the gain at the Nyquist frequency) to blur and downsample Ref. To generate the PAN with the size of $320 \times 320$, we use the PAN response of the IKONOS-like reflection spectrum response filter to sample Ref along with the spectral mode. Similarly, we simulate datasets B and C. The size of Ref on dataset B is $208 \times 208 \times 93$ and that on dataset C is $320 \times 320 \times 93$. On these datasets, the resolution ratio between PAN and HSI is 16 and that between PAN and MSI is 4.

Dataset D is a real dataset generated from the HSI captured by GF-5 and GF-1 satellites launched in China. GF-5 satellite includes a hyperspectral imager with a spatial resolution of 30 m and a spectral resolution of 5–10 nm with 330 bands. GF-1 satellite provides MS and PAN with spatial resolutions of 8 and 2 m, respectively. The sizes of the test HSI, MSI, and PAN in the real experiment are $32 \times 32 \times 246$, $128 \times 128 \times 4$, and $480 \times 480$, respectively (to satisfy the requirement of integral resolution ratio between HSI and MSI in some compared methods such as CNMF, we slightly scale the resolution of MSI from $120 \times 120 \times 4$ to $128 \times 128 \times 4$).

We compare HyperFusion with a variety of representative methods, including FHM methods: HSI super-resolution via subspace-based regularization named HySure [24], GSA based on the synthesized band denoted as GSA (M) [67], hyperspectral super-resolution by coupled spectral unmixing (HSRCSU) [68], CNMF [31], and learning a low tensor-train rank representation (LTTR) for HSI super-resolution [36]; FHP methods: GLP [44] and GSA for FHP represented as GSA (P) [28], [29]; and FHMP methods: hyperspectral, multispectral, and PAN data fusion based on CNMF denoted as CNMFFHMP [55] and fusing multiple multiband images (FMMIs) [57]. To demonstrate the superiority of HyperFusion, a two-step fusion strategy is also compared. This two-step fusion strategy can be considered as a special FHMP method, which first obtains a fused high-resolution MSI by fusing MSI and PAN and then generates a fused high-resolution HSI by fusing HSI and the fused MSI. The combination of GLP (first step: fusion of MSI and PAN) and GSA (M) (second step: FHM) (denoted as GLPGSA) is selected for comparison, which achieves satisfactory fusion performance in our experiment. The upsampled HSI by image interpolation with a polynomial kernel with 23 coefficients [69] (denoted as EXP) is also provided as a yardstick. The abbreviations of the compared methods are summarized in Table I.

We select eight typical indices [22], [28], [70], [71] to quantitatively evaluate the fusion performance of different methods in the simulation experiment, including relative dimensionless global error in synthesis (ERGAS), spectral angle mapper (SAM), relative average spectral error (RASE), root-mean-square error (RMSE), mean universal image quality index (QAVG), correlation coefficient (CC), spatial correlation coefficient (SCC), and mean structural similarity (MSSIM). ERGAS evaluates the overall performance of all spectral bands of the fusion result. SAM represents the spectral distortion by the absolute angles between the two vectors constructed from each pixel of the fusion and the reference images. RASE is used to estimate the global spectral quality of the fused images. RMSE is used to calculate the standard deviation between the fusion image and the reference image. QAVG is a metric for the degree of structural distortion. CC reflects the similarity

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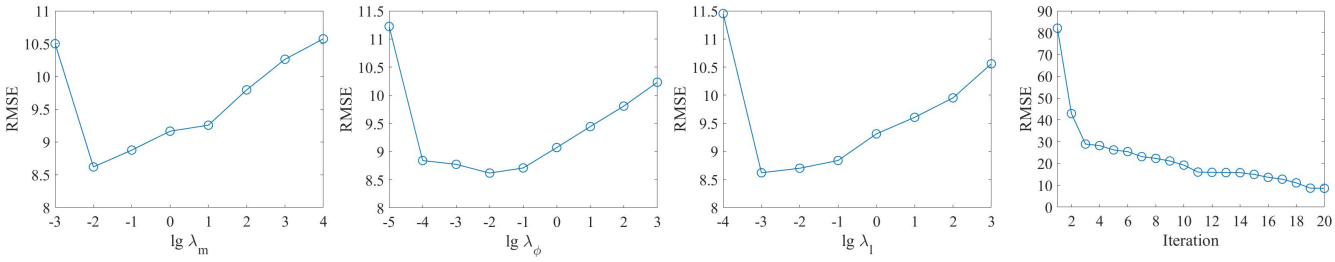



Fig. 3. Parameter setting and convergence analysis. (a) Sensitivity analysis for parameter $\lambda_m$. (b) Sensitivity analysis for parameter $\lambda_\phi$. (c) Sensitivity analysis for parameter $\lambda_l$. (d) Convergence of HyperFusion.

TABLE I
ABBREVIATIONS OF THE COMPARED METHODS

| Abbreviations | Meanings |
|---|---|
| HySure [24] | Hyperspectral image super-resolution via subspace-based regularization. |
| GSA (M) [67] | Gram-Schmidt adaptive based on the synthesized band. |
| HSRCSU [68] | Hyperspectral super-resolution by coupled spectral unmixing. |
| CNMF [31] | Coupled non-negative matrix factorization. |
| LTTR [36] | Learning a low tensor-train rank representation for hyperspectral image super-resolution. |
| GLP [44] | Generalized Laplacian pyramid. |
| GSA (P) [28], [29] | Gram-Schmidt adaptive for FHP. |
| CNMFFHMP [55] | Hyperspectral, multispectral, and panchromatic data fusion based on coupled non-negative matrix factorization. |
| FMMI [57] | Fusing multiple multiband images. |
| GLPGSA | Combination of GLP (first step: fusion of MSI and PAN) and GSA (M) (second step: FHM). |
| EXP [69] | Image interpolation with a polynomial kernel with 23 coefficients. |

degree of spectral features. SCC indicates the degree of the spatial detail information of one image presented in the other one. MSSIM is an index measuring the MSSIM between two images. In the real experiment, we utilize the no-reference HSI quality metric (NHQM) and the spectral score (SS) [72], [73] for quantitative analysis.

For compared methods, we set their parameters to those of the best performance on all datasets. We set $\mu = 0.001$ as an empirical value in our experiment. $\lambda_m$, $\lambda_\phi$, and $\lambda_l$ are important parameters to balance the importance of different terms. We analyze the sensitivities of $\lambda_m$, $\lambda_\phi$, and $\lambda_l$ for HyperFusion through the following experiment by fixing other parameters. We select dataset A as an example for analysis, and RMSE is adopted for quantitative evaluation. The results are shown in Fig. 3. From the results, RMSEs decrease with the increase of $\lambda_m$, $\lambda_\phi$, and $\lambda_l$ when $\lg \lambda_m < -2$, $\lg \lambda_\phi < -2$, and $\lg \lambda_l < -3$, respectively. When $\lambda_m$, $\lambda_\phi$, and $\lambda_l$ grow further after $\lg \lambda_m > -2$, $\lg \lambda_\phi > -2$, and $\lg \lambda_l > -3$, RMSEs gradually increase. Therefore, we choose $\lambda_m = 0.01$, $\lambda_\phi = 0.01$, and $\lambda_l = 0.001$ in our experiment. It is difficult to prove the convergence of HyperFusion theoretically because the optimization problem of HyperFusion is not strong convex. It is widely acknowledged that the alternative direction multiplier method performs well in many applications and converges to a local optimum result when proper parameters are selected. Then, we adopt an empirical analysis of convergence for HyperFusion, as shown in Fig. 3(d). It demonstrates that HyperFusion has a rapid rate of convergence evaluated by RMSE.

### *B. Simulation Experiment*

In the simulation experiment, we compare the fusion results of different methods in terms of visual quality and quantitative analysis. For visual comparison, we randomly select three bands of the fusion result to form a false-color image for demonstration and the residual image by the mean absolute errors between Ref and the fusion result for comparison.

*1) Dataset A:* The visual comparison of different methods on dataset A is shown in Fig. 4. The 70th, 40th, and 10th bands are selected to generate the false-color image, and the regions labeled by small boxes are enlarged as large boxes in the left and lower parts of Fig. 4. From Fig. 4, the overall spatial quality of fusion result in LTTR is more blurry than other methods. For example, the edges of the blue building in the middle of Fig. 4(e) are not clear. HSRCSU in Fig. 4(c) can provide better spatial clarity than LTTR, and however, the color of HSRCSU is slightly different from that of Ref in Fig. 4(n), exhibiting some spectral distortions. This phenomenon can be observed from the enlargement in Fig. 4(c), whose color is a little darker than that of Ref. Compared with FHM methods such as HySure, GSA (M), HSRCSU, CNMF, and LTTR, FHP methods, such as GLP and GSA (P) in Fig. 4(f) and (g), are excellent in the sharpness of spatial details by introducing PANs into the fusion process. However, there are obvious spectral distortions in GLP and GSA (P). For example, the colors of the buildings in enlargements of Fig. 4(f) and (g) are too dark or bright, different from that of Ref in Fig. 4(n). Since CNMFFHMP, FMMI, GLPGSA, and HyperFusion effectively fuse the information of HSI, MSI, and PAN, CNMFFHMP, FMMI, GLPGSA, and HyperFusion in Fig. 4(h)–(k) demonstrate more spatial details compared with HySure, GSA (M), HSRCSU, CNMF, and LTTR while keeping better spectral information than GLP and GSA (P). Due to the inaccurate estimation of spectral response transform vector between PAN and HSI, CNMFFHMP and FMMI still have some spectral distortions (e.g., the colors of the enlargements in Fig. 4(h) and (i) are different from that of Ref). GLPGSA has smaller spectral distortion than CNMFFHMP and FMMI,

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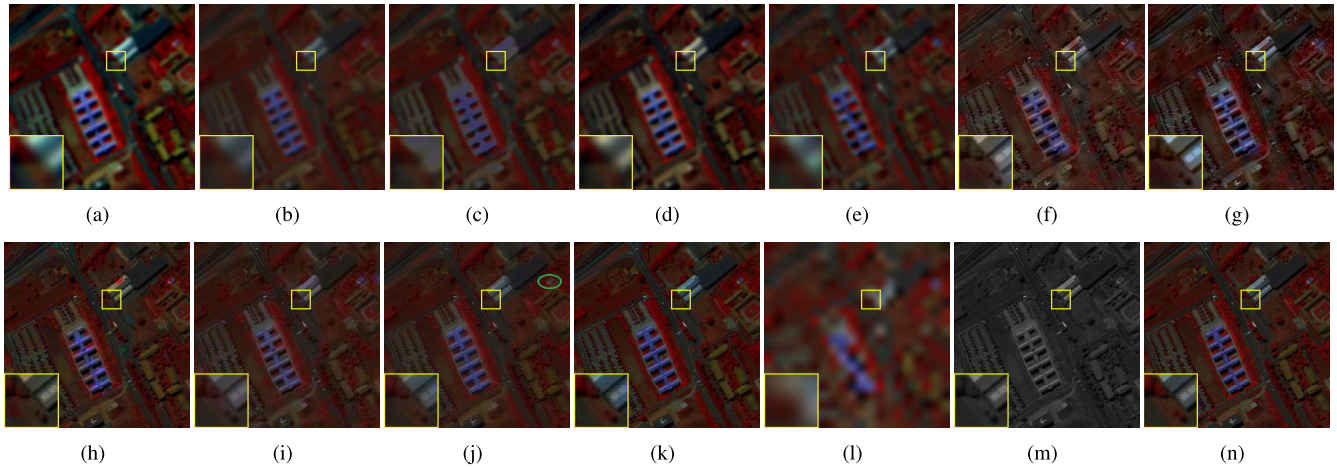



Fig. 4. Visual comparison of results on the simulated dataset A. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP. (m) PAN. (n) Ref. Large boxes are enlargements of the small boxes. All images, except (m), are in false color (bands 70, 40, and 10).

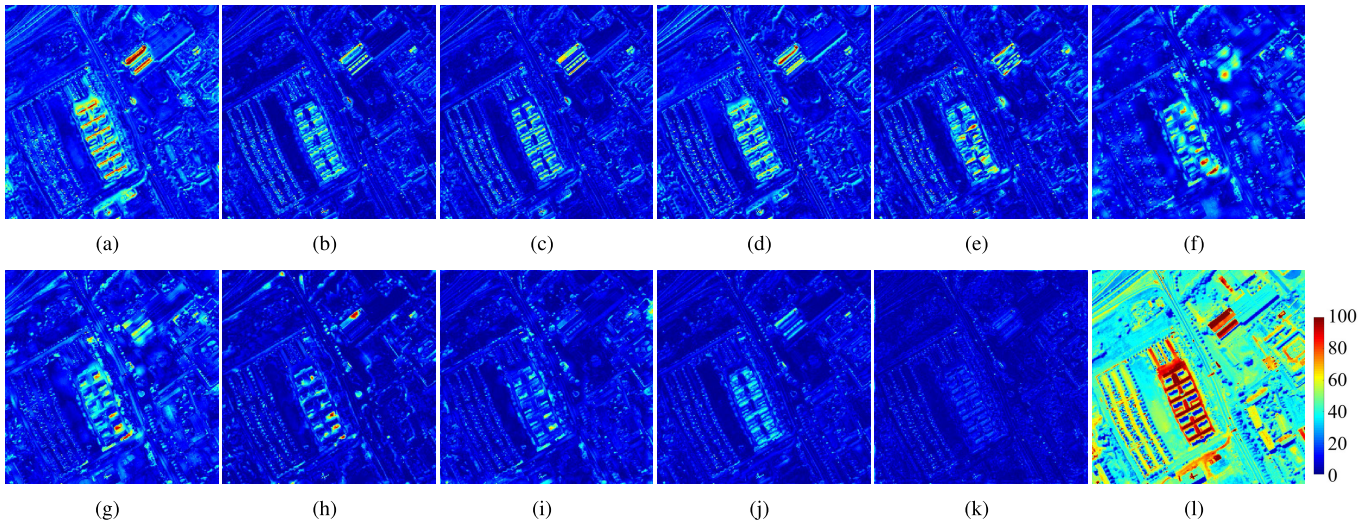



Fig. 5. Residual images of fusion results in the fusion experiment on the simulated dataset A. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP.

and however, its result is a little blurrier than HyperFusion [e.g., the clarity of the region labeled by a green ellipse in Fig. 4(j) is lower than that in Fig. 4(k)] (this phenomenon can be clearly observed from the residual images in the following). Therefore, HyperFusion is closest to Ref among all methods, which has the best balance between the spatial and spectral quality. To further indicate the above conclusion, we calculate the residual images by the mean absolute errors between fusion results and Ref. The experimental results are shown in Fig. 5. As shown in Fig. 5(a), (b), (d), and (e), large errors are mainly along edges in the residual images of HySure, GSA (M), CNMF, and LTTR because fusion results of these methods are blurry. Caused by significant spectral distortions, there are some relatively large values in many regions of Fig. 5(f) and (g) in GLP and GSA (P). Due to some slight spectral distortions in CNMFFHMP and FMMI, Fig. 5(h) and (i) still have some large values. While there are still some blurry regions in GLPGSA, some large errors along edges exist in Fig. 5(j). Conversely, HyperFusion in Fig. 5(k) only has a few small residuals at the edges of the buildings and the smallest residuals among all methods, indicating its superiority over other state of the arts.

*2) Dataset B:* In this section, we verify the effectiveness of HyperFusion by using dataset B. Different from the images on dataset A mainly composed of urban buildings, the images on dataset B include targets of farmlands. The visual comparison of different methods on this dataset is shown in Fig. 6. The 61st, 42th, and 20th bands are selected to generate the false-color image. There are some false stripes in Fig. 6(e), degrading the fusion performance of LTTR. Fusion results of HySure, GSA (M), HSRCSU, and CNMF in Fig. 6(a)–(d) are similar, and their spatial details are blurry. Compared with the above-mentioned methods, GLP and GSA (P) in Fig. 6(f) and (g) have better clarity of spatial details benefiting from the fusion of the PAN of a high spatial resolution. However, some spatial artifacts (e.g., dark spots) exist in GLP and GSA (P), influencing their visual quality. The colors of the enlargements in Fig. 6(h) and (i) are slightly different from that

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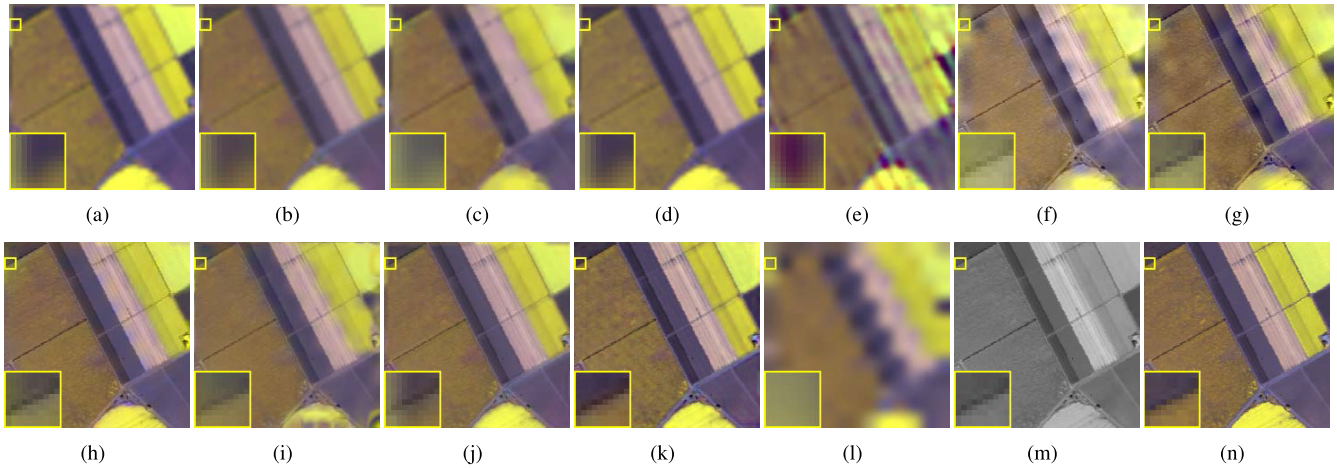



Fig. 6. Visual comparison of fusion results on the simulated dataset B. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP. (m) PAN. (n) Ref. All images, except (m), are in false color (bands 61, 42, and 20).

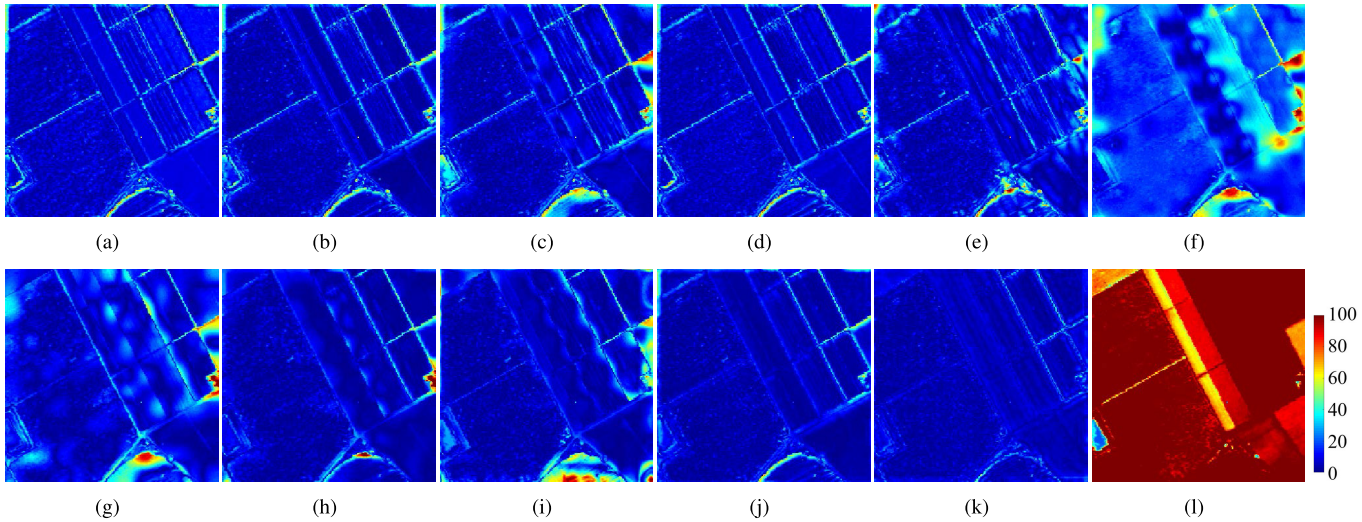



Fig. 7. Residual images of fusion results in the fusion experiment on the simulated dataset B. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP.

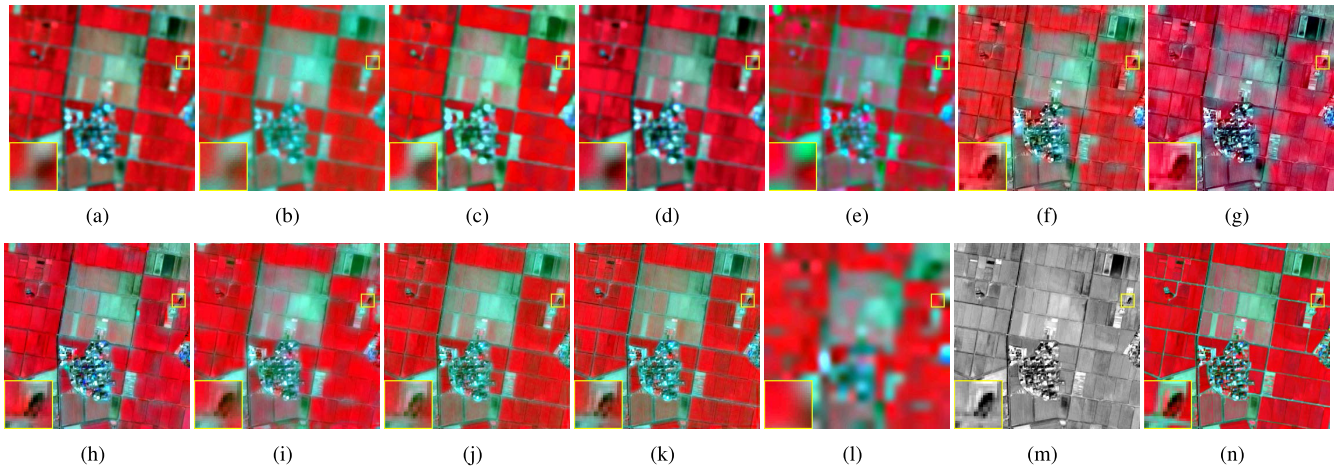



Fig. 8. Visual comparison of fusion results on the simulated dataset C. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP. (m) PAN. (n) Ref. Large boxes are enlargements of the small boxes. All images, except (m), are in false color (bands 61, 42, and 20).

of Ref in Fig. 6(n), demonstrating some spectral distortions in CNMFFHMP and FMMI. The enlargement of GLPGSA in Fig. 6(j) has smaller spectral distortions than CNMFFHMP and FMMI, but it is a little blurry and has some false stripes. HyperFusion has the best balance between spatial and spectral quality among all methods. For example, its color in Fig. 6(k)

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is close to the Ref in Fig. 6(n), while clear spatial details can be provided. Fig. 7 shows the comparison of residual images on dataset B. For HySure, GSA (M), HSRCSU, CNMF, and LTTR, the edge regions of residual images always have large values, as shown in Fig. 7(a)–(e). Significant spectral distortions exist in GLP and GSA (P), leading to large values in several regions of Fig. 7(f) and (g). From the comparison of residual images in Fig. 7, the residuals of HyperFusion are the smallest among all methods, indicating excellent fusion performance of HyperFusion.

*3) Dataset C:* To further verify the effectiveness of our method for data with different platforms, dataset C captured from an airborne flight is adopted in this experiment. The visual comparison of fusion results on dataset C is shown in Fig. 8. The 61st, 42th, and 20th bands are selected to generate the false-color image. There are slight spectral distortions of HySure in Fig. 8(a), while its spatial quality is also unsatisfactory. From the comparison of enlargements in Fig. 8(b)–(e), GSA (M), HSRCSU, and CNMF provide clearer spatial details than that of LTTR. Compared with other methods, GLP and GSA (P) in Fig. 8(f) and (g) generally have the clearest spatial clarity. However, their spectral distortions are also obvious. For example, the color of farmlands in the middle of Fig. 8(g) is close to blue, while that of Ref in Fig. 8(n) is light green. Due to the spectral distortion of CNMFFHMP, FMMI, and GLPGSA, there are some abnormal red values in the enlargements of Fig. 8(h) and (i), and the color in the middle of enlargement in Fig. 8(j) tends to be green. Also, the enlargement in Fig. 8(j) is blurrier than HyperFusion. As shown in Fig. 8(k), HyperFusion demonstrates satisfactory spatial quality while efficiently preserving the spectral information, achieving the best fusion quality among all methods. This conclusion can be further verified from residual images in Fig. 9, where we calculate the residual images between fusion results and Ref. We can find that HyperFusion has the smallest error among all methods.

*4) Quantitative Analysis:* The quantitative results on dataset A are listed in Table II. The last row of the table represents the ideal value of each index, and the values shown in the green and blue colors of the table represent the best and second best results of the corresponding indices, respectively. Since the spatial details of FHM, such as HySure, GSA (M), HSRCSU, CNMF, and LTTR, are always blurry, the values of QAVG of these methods are not high (e.g., QAVG$<$ 0.65). GLP and GSA (P) can solve the blur problem based on the FHP. However, the existed spectral distortions of GLP and GSA (P) lead to large values of SAM (e.g., SAM$>$ 10). CNMFFHMP and FMMI suffer some slight spectral distortions, and therefore, their SAM is not very small. GLPGSA suffers less spectral distortions than CNMFFHMP and FMMI, but its spatial quality is lower than CNMFFHMP. Therefore, the QAVG of GLPGSA is lower than that of CNMFFHMP. By simultaneous FHMP, HyperFusion constructs an efficient fusion model that provides clear spatial details while keeping the spectral information well, leading to the closest to Ref. As a result, HyperFusion has the best performance of all metrics on dataset A. Similar conclusions can be made from the quantitative results on datasets B and C. These results are listed in Tables III and IV. Similar to Table II, the values of SAM of GLP and GSA (P) are always large in Tables III and IV caused by spectral distortions. Consequently, the values of the overall metric ERGAS of GLP and GSA (P) are inferior to those of most fusion methods. HyperFusion achieves the best performance of four and eight metrics in Tables III and IV, respectively. Therefore, we can conclude the effectiveness of HyperFusion in quantitative analysis.

TABLE II

COMPARISON OF QUANTITATIVE INDICES IN THE FUSION EXPERIMENT ON SIMULATED DATASET A. **GREEN**: THE BEST. **BLUE**: THE SECOND BEST

| Method | ERGAS | SAM | RASE | RMSE | QAVG | CC | SCC | MSSIM |
|---|---|---|---|---|---|---|---|---|
| HySure | 3.395 | 13.768 | 43.408 | 22.828 | 0.636 | 0.855 | 0.440 | 0.551 |
| GSA (M) | 2.261 | **7.602** | 29.447 | 15.135 | 0.566 | 0.832 | 0.441 | 0.649 |
| HSRCSU | 2.293 | 7.733 | 28.632 | 15.600 | 0.621 | 0.844 | 0.420 | 0.649 |
| CNMF | 2.664 | 8.996 | 31.399 | 18.649 | 0.645 | 0.849 | 0.438 | 0.623 |
| LTTR | 2.423 | 8.386 | 32.246 | 16.230 | 0.591 | 0.800 | 0.421 | 0.641 |
| GLP | 2.780 | 12.233 | 41.899 | 18.291 | 0.682 | 0.760 | 0.801 | 0.771 |
| GSA (P) | 2.754 | 14.183 | 43.896 | 17.733 | 0.681 | 0.799 | 0.779 | 0.730 |
| CNMFFHMP | 1.796 | 9.486 | 28.646 | 12.349 | **0.817** | 0.855 | **0.844** | **0.844** |
| FMMI | 1.777 | 9.831 | 25.991 | 12.462 | 0.770 | 0.880 | 0.807 | 0.804 |
| GLPGSA | **1.717** | 7.683 | **24.323** | **11.512** | 0.789 | **0.889** | 0.816 | 0.816 |
| HyperFusion | **1.282** | **7.531** | **19.670** | **8.619** | **0.904** | **0.945** | **0.867** | **0.849** |
| EXP | 3.494 | 12.072 | 47.512 | 22.819 | 0.166 | 0.521 | 0.188 | 0.548 |
| Ideal Value | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

TABLE III

COMPARISON OF QUANTITATIVE INDICES IN THE FUSION EXPERIMENT ON SIMULATED DATASET B. **GREEN**: THE BEST. **BLUE**: THE SECOND BEST

| Method | ERGAS | SAM | RASE | RMSE | QAVG | CC | SCC | MSSIM |
|---|---|---|---|---|---|---|---|---|
| HySure | 0.757 | 4.118 | 15.951 | 14.350 | 0.709 | 0.725 | 0.809 | 0.780 |
| GSA (M) | 0.655 | **3.069** | **13.217** | 12.483 | 0.684 | **0.750** | 0.812 | 0.790 |
| HSRCSU | 0.850 | 3.871 | **13.438** | 15.973 | 0.623 | **0.743** | 0.802 | 0.773 |
| CNMF | 0.701 | 3.592 | 17.160 | 13.505 | 0.699 | 0.697 | 0.809 | 0.786 |
| LTTR | 0.793 | 3.885 | 17.491 | 15.171 | 0.633 | 0.629 | 0.798 | 0.764 |
| GLP | 1.478 | 5.749 | 22.899 | 26.377 | 0.634 | 0.625 | 0.874 | 0.830 |
| GSA (P) | 0.987 | 4.756 | 14.504 | 17.371 | 0.654 | 0.699 | 0.906 | 0.870 |
| CNMFFHMP | 0.748 | 4.106 | 20.569 | 12.512 | 0.787 | 0.530 | **0.913** | **0.908** |
| FMMI | 0.864 | 4.299 | 14.650 | 15.402 | 0.727 | 0.684 | 0.879 | 0.852 |
| GLPGSA | **0.564** | **3.128** | 14.267 | **10.321** | **0.813** | 0.708 | 0.901 | **0.886** |
| HyperFusion | **0.509** | 3.501 | 18.698 | **9.128** | **0.837** | 0.630 | **0.914** | 0.885 |
| EXP | 1.226 | 4.978 | 14.005 | 23.523 | 0.396 | 0.717 | 0.750 | 0.721 |
| Ideal Value | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

## C. Real Experiment

In this section, we adopt the real dataset D to verify the efficiency of HyperFusion. The visual comparison of different fusion methods on the real dataset D is shown in Fig. 10. The 200th, 100th, and 10th bands are selected to generate the false-color image. There are some spatial artifacts in LTTR of Fig. 10(e). Compared with HySure, GSA (M), HSRCSU, CNMF in Fig. 10(a)–(d), GLP, GSA (P), CNMFFHMP, FMMI, GLPGSA, and HyperFusion have better spatial clarity. However, some spectral information is lost in GLP, GSA (P), and CNMFFHMP, influencing their fusion performance. For example, the colors of the lines in the center regions of Fig. 10(f)–(h) are different from that

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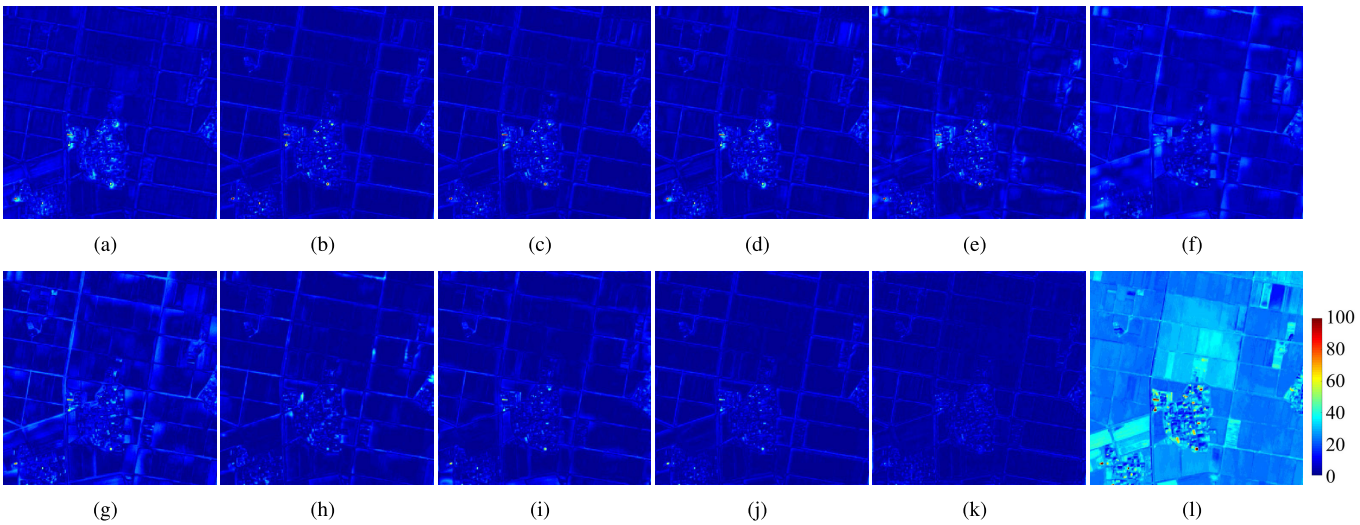



Fig. 9. Residual images of fusion results in the fusion experiment on the simulated dataset C. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP.

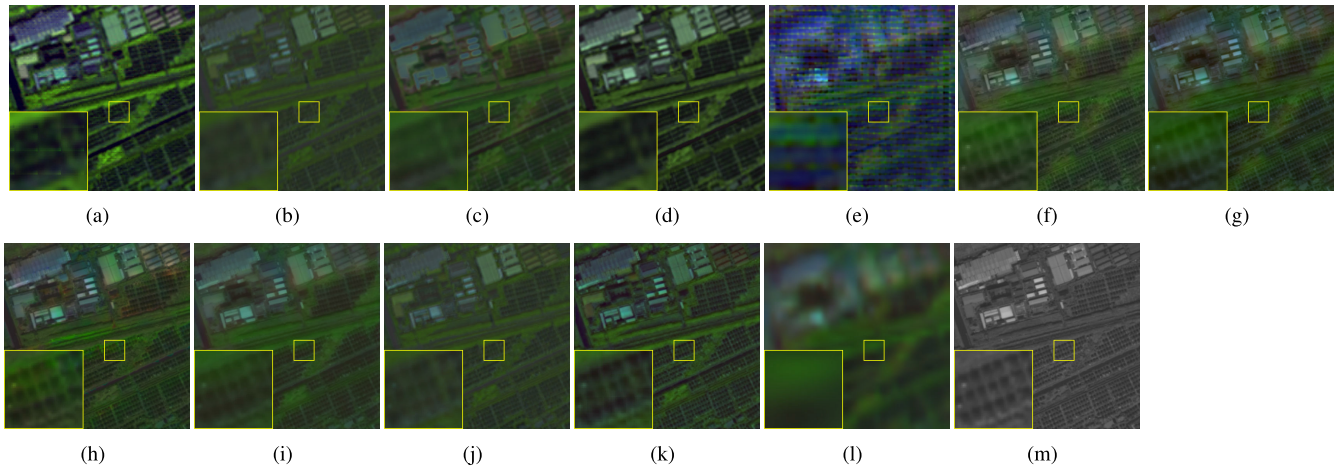



Fig. 10. Visual comparison of results on the real dataset D. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP. (m) PAN. Large boxes are enlargements of the small boxes. All images, except (m), are in false color (bands 200, 100, and 10).

TABLE IV

COMPARISON OF QUANTITATIVE INDICES IN THE FUSION EXPERIMENT ON SIMULATED DATASET C. **GREEN**: THE BEST. **BLUE**: THE SECOND BEST

| Method | ERGAS | SAM | RASE | RMSE | QAVG | CC | SCC | MSSIM |
|---|---|---|---|---|---|---|---|---|
| HySure | 2.371 | 8.546 | 22.464 | 7.507 | 0.672 | 0.833 | 0.540 | 0.848 |
| GSA (M) | 1.768 | 5.603 | 15.668 | 5.795 | 0.614 | 0.827 | 0.519 | 0.893 |
| HSRCSU | 1.812 | 5.561 | 16.805 | 5.836 | 0.636 | 0.797 | 0.531 | 0.897 |
| CNMF | 1.874 | 5.684 | 16.643 | 6.191 | 0.665 | 0.816 | 0.534 | 0.890 |
| LTTR | 2.091 | 7.298 | 16.902 | 6.791 | 0.620 | 0.794 | 0.504 | 0.865 |
| GLP | 2.872 | 10.822 | 26.858 | 9.311 | 0.387 | 0.674 | 0.644 | 0.838 |
| GSA (P) | 2.804 | 11.992 | 21.863 | 9.003 | 0.387 | 0.695 | 0.624 | 0.810 |
| CNMFFHMP | 1.857 | 7.138 | 17.405 | 5.996 | 0.654 | 0.794 | **0.725** | 0.906 |
| FMMI | 1.834 | 6.813 | 16.544 | 5.907 | 0.634 | 0.807 | 0.669 | 0.900 |
| GLPGSA | **1.517** | **5.531** | **13.620** | **4.894** | **0.717** | **0.873** | 0.716 | **0.916** |
| HyperFusion | **1.344** | **5.353** | **12.519** | **4.321** | **0.828** | **0.917** | **0.771** | **0.924** |
| EXP | 2.850 | 9.538 | 24.134 | 9.352 | 0.226 | 0.559 | 0.355 | 0.836 |
| Ideal Value | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

of FHM methods, FMMI, and HyperFusion. Although FMMI and GLPGSA suffer small spectral distortions, their results are slightly blurry. For example, the grids in the regions of enlargement in Fig. 10(i) and (j) are not clear. Conversely, HyperFusion in Fig. 10(k) provides clear spatial details and excellent spectral quality, indicating its effectiveness on the real dataset D. This conclusion can be further verified from the quantitative results in Table V, where HyperFusion achieves the smallest NHQM and second smallest SS in all methods.

### *D. Analysis of Computational Efficiency*

To compare the computational efficiency of different methods, the computational time (in seconds) of comparable methods on different datasets is shown in Table VI. All methods are run in MATLAB 2016a on a 2.81-GHz computer with an Intel Core i5-8400 CPU and 8-GB RAM. We notice that for most methods, the computational time scales about linearly with the size of the fused HSI. From Table VI, GSA (M) [or GSA (P)] is the most efficient method, which requires the smallest calculation time. Due to the iterative process for solving optimization models, a large amount of time is required in HySure, CNMF, LTTR, FMMI, and HyperFusion on all datasets. Benefiting from the calculation process on the subspace matrix, the computational cost of HyperFusion

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TABLE V

COMPARISON OF QUANTITATIVE INDICES IN THE FUSION EXPERIMENT ON THE REAL DATA. GREEN: THE BEST. BLUE: SECOND BEST

| Method | HySure | GSA (M) | HSRCSU | CNMF | LTTR | GLP | GSA (P) | CNMFFHMP | FMMI | GLPGSA | HyperFusion | EXP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NHQM | 27.115 | 27.438 | 24.624 | 26.590 | 29.099 | 18.666 | **18.322** | 18.378 | 18.906 | 21.239 | **18.259** | 52.344 |
| SS | 0.834 | 0.867 | 0.850 | **0.802** | 0.986 | 0.925 | 0.931 | 0.904 | 0.830 | 0.888 | **0.814** | 0.930 |

TABLE VI

COMPARISON OF COMPUTATIONAL TIME OF EACH METHOD. UNIT: SECOND

| Data name | dataset A | dataset B | dataset C | dataset D |
|---|---|---|---|---|
| HySure | 9.78 | 3.92 | 9.87 | 23.29 |
| GSA (M) | 0.83 | 0.41 | 0.80 | 5.48 |
| HSRCSU | 3.88 | 2.46 | 6.34 | 18.46 |
| CNMF | 8.06 | 3.79 | 8.15 | 17.28 |
| GLP | 2.83 | 1.58 | 2.81 | 14.17 |
| LTTR | 14.92 | 6.78 | 13.52 | 89.17 |
| GSA (P) | 0.67 | 0.29 | 0.67 | 9.29 |
| CNMFFHMP | 2.07 | 1.03 | 2.04 | 8.51 |
| FMMI | 17.12 | 6.99 | 17.10 | 42.46 |
| GLPGSA | 1.91 | 0.68 | 1.23 | 17.23 |
| HyperFusion | 16.61 | 9.16 | 15.84 | 43.11 |
| EXP | 0.371 | 0.306 | 0.374 | 2.843 |

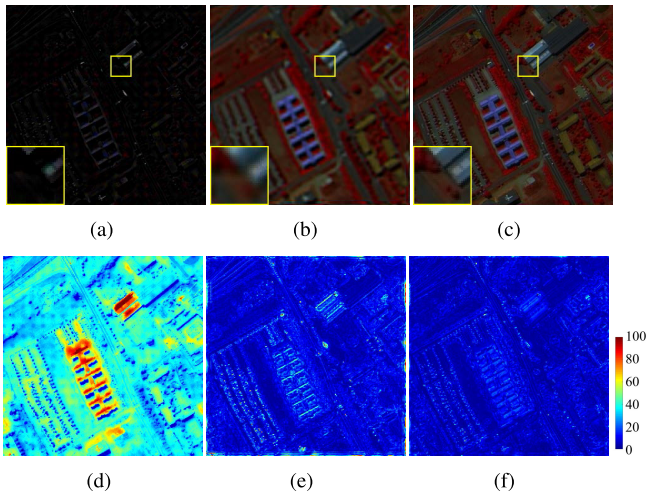



Fig. 11. Ablation experiment on simulated dataset A. (a) HyperFusion without MSI. (b) HyperFusion without PAN. (c) HyperFusion. (d) Residual image of (a). (e) Residual image of (b). (f) Residual image of (c).

is comparable to HySure, CNMF, LTTR, and FMMI. The time-consuming problem of HyperFusion is solvable when the GPU acceleration is adopted.

### *E. Ablation Study*

To investigate the effect of the proposed method by simultaneous FHMP, we ignore the data fidelity term $\|\mathbf{M} - \mathbf{R}_m\mathbf{E}\mathbf{X}\mathbf{B}_m\mathbf{S}_m\|_F^2$ and $\| \bigtriangledown \tilde{\mathbf{P}} - \bigtriangledown\mathbf{R}_m\mathbf{E}\mathbf{X}\|_{2,1}$ of (7). The experimental results are shown in Fig. 11. Compared with HyperFusion without PAN in Fig. 11(b), HyperFusion in Fig. 11(c) has clearer spatial details and provides better spectral information than HyperFusion without MSI in Fig. 11(a). Therefore, we can conclude that simultaneous FHMP is necessary and important for HyperFusion. This conclusion

TABLE VII

COMPARISON OF QUANTITATIVE INDICES FOR THE ABLATION EXPERIMENT ON SIMULATED DATASET A. GREEN: THE BEST. BLUE: THE SECOND BEST

| Method | ERGAS | SAM | RASE | RMSE | QAVG | CC | SCC | MSSIM |
|---|---|---|---|---|---|---|---|---|
| HyperFusion w/o MSI | 6.103 | 24.802 | 65.673 | 46.516 | 0.134 | 0.649 | **0.675** | 0.195 |
| HyperFusion w/o PAN | **1.930** | **8.153** | **25.116** | **13.156** | **0.790** | **0.910** | 0.541 | **0.674** |
| HyperFusion | **1.282** | **7.531** | **19.670** | **8.619** | **0.904** | **0.945** | **0.867** | **0.849** |
| Ideal Value | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

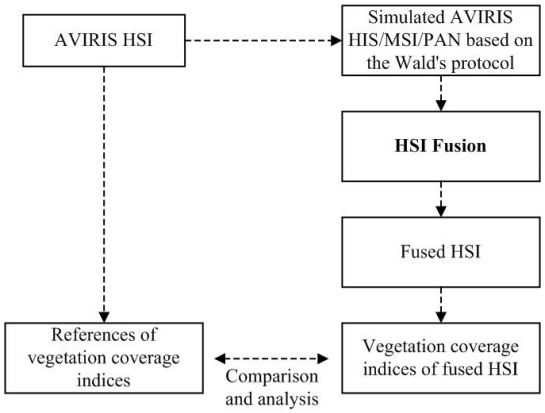



Fig. 12. Experimental flow of vegetation index analysis of HSI fusion results.

can also be verified from the comparison of residuals in Fig. 11(d)–(f), where HyperFusion has the smallest error. The comparison of quantitative indices for the ablation experiment on dataset A is listed in Table VII. HyperFusion has the best fusion performance. For example, it has the best values of eight metrics, demonstrating that HSI, MSI, and PAN are all essential elements for HyperFusion.

## V. EXPERIMENT OF VEGETATION COVERAGE INDEX ANALYSIS

In the field of hyperspectral remote sensing monitoring, the extraction of land cover types and the analysis of changes are of great significance for the implementation of precision agriculture. Due to unique spectral absorption characteristics of different ground features and abundant resources provided by remote sensing data for measuring land cover parameters, various vegetation indices have been proposed for the analysis of vegetation coverage. To demonstrate the effectiveness of HyperFusion in the field of remote sensing, we carry out the following experiment of vegetation coverage index analysis in this section.

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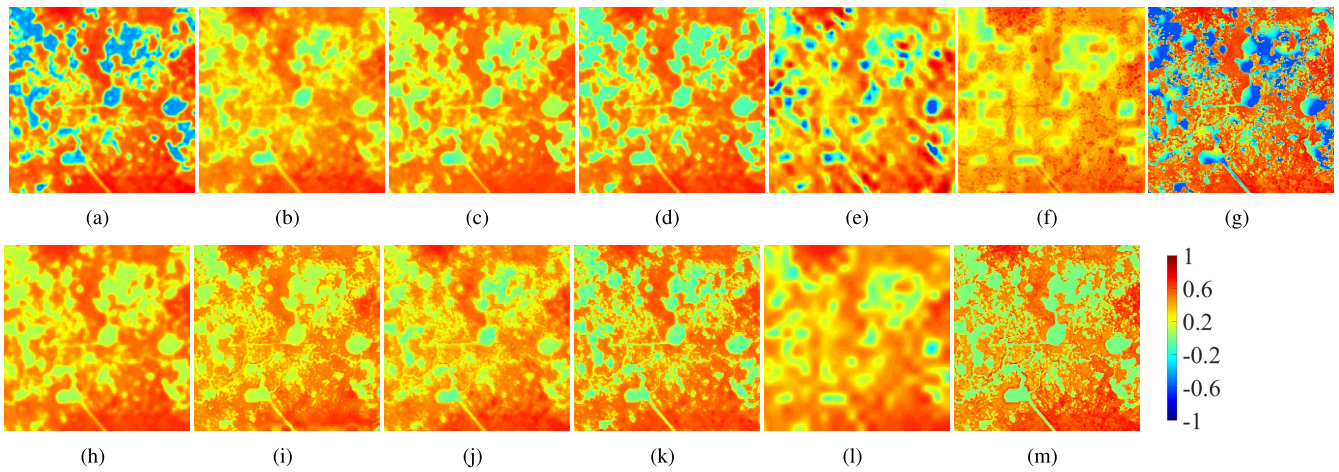



Fig. 13. Fused NDVI visual comparison of different methods. (a) HySure. (b) GSA (M). (c) HSRCSU. (d) CNMF. (e) LTTR. (f) GLP. (g) GSA (P). (h) CNMFFHMP. (i) FMMI. (j) GLPGSA. (k) HyperFusion. (l) EXP. (m) Ref.

TABLE VIII
RMSE COMPARISON OF VEGETATION COVERAGE INDICES OF FUSION IMAGES OF VARIOUS METHODS. GREEN: BEST

| Method | HySure | GSA (M) | HSRCSU | CNMF | LTTR | GLP | GSA (P) | CNMFFHMP | FMMI | GLPGSA | HyperFusion | EXP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NDVI | 0.282 | 0.193 | 0.180 | 0.188 | 0.259 | 0.293 | 0.305 | 0.186 | 0.144 | 0.155 | **0.094** | 0.262 |
| NDII | 0.228 | 0.225 | 0.213 | 0.215 | 0.378 | 0.243 | 0.163 | 0.215 | 0.160 | 0.134 | **0.096** | 0.327 |
| SIWSI | 0.203 | 0.214 | 0.209 | 0.207 | 0.477 | 0.190 | 0.140 | 0.208 | 0.158 | 0.125 | **0.098** | 0.307 |
| DSWI | 0.283 | 0.285 | 0.294 | 0.288 | 0.390 | 0.401 | 0.210 | 0.292 | 0.215 | **0.188** | **0.188** | 0.422 |
| $LWVI_1$ | 0.240 | 0.158 | 0.146 | 0.151 | 0.171 | 0.189 | 0.263 | 0.145 | 0.129 | 0.114 | **0.096** | 0.219 |

The simulated HSI comes from the AVIRIS sensor, and the experimental process is shown in Fig. 12. We selected five indices as examples to compare the performance of different fusion methods. The selected indices include NDVI [74], [75], NDII [76], SIWSI [77], DSWI [78], and $LWVI_1$ [78]. In the study of vegetation coverage, NDVI uses the near-infrared band and visible light red band of remote sensing images. NDII is used for rice extraction and grassland degradation research. SIWSI has been constructed to detect drought conditions on the ground. DSWI and $LWVI_1$ are often used to detect changes in climate conditions, especially the impact of drought on ecosystems.

We select the NDVI as an example for visual comparison of different methods, as shown in Fig. 13. Generally speaking, HyperFusion has the best fusion results evaluated from the aspects of spatial quality and consistency to Ref. We further utilize RMSE to quantitatively compare the results of vegetation coverage indices calculated from different fusion methods. The experimental results are shown in Table VIII. It demonstrates that HyperFusion has the smallest RMSE in all indices, which are closest to reference values among all comparable methods. Therefore, we can conclude that HyperFusion will be efficient in the field of remote sensing.

## VI. CONCLUSION

To generate an HSI with a high spatial resolution, we propose a computational approach for hyperspectral, multispectral, and PAN image fusion, named HyperFusion. In particular, we establish two data fidelity terms to preserve the spatial and spectral information of HSI and MSI. To efficiently transfer the spatial details of the PAN image into the fused HSI while keeping the spectral information well, we further construct a prior constraint from PAN based on the structural similarity. To accurately describe latent characteristics of the fused HSI, we construct another low-rank prior constraint on the coefficient matrix decomposed from the fused HSI. By incorporating the above data fidelity terms and prior constraints, we formulate the HSI, MSI, and PAN fusion as an optimization problem, which is solved by the alternative direction multiplier method efficiently. In the experiment, we first utilize three simulated datasets from ROSIS, AVIRIS sensors, and an airborne flight platform to verify the effectiveness of HyperFusion. From the aspects of visual comparison and quantitative analysis on these simulated datasets, we can find that HyperFusion provides satisfactory spatial details while keeping the spectral information well. As a result, HyperFusion achieves the best balance between spatial and spectral quality compared with other methods. More importantly, the superiority of HyperFusion over other state of the arts in a real application is further verified by a real experiment based on the fusion of images from GF-5 and GF-1 satellites. Meanwhile, the effectiveness of HyperFusion in remote sensing is also demonstrated by a simulated vegetation coverage experiment.

In the future, the following studies are our next work.

1) HSI, MSI, and PAN should be well registered [79] before fusion because HyperFusion is not robust to misalignments between the source imageries. Therefore,

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image matching algorithms will be studied and incorporated into the fusion process if images from different satellites are unregistered.

2) Design an efficient interpretable deep CNN [80] to further improve the performance of HyperFusion which is also one of our interests.
3) More real experiments from different satellites will be further investigated.

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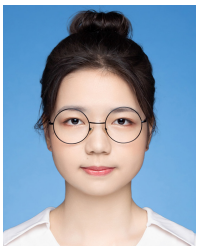
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